



# Yuxuan Liu

Ph.D. Candidate in Trustworthy Generative Audio and Music AI ·  
Expected to Graduate Mar. 2027

AI Algorithms · Audio  
Intelligence  
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Google Scholar

## Education

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|--------------------------------------------------------------------------------------------------------|-------------------------|----------------------------------------------------------------|---------------|
| Beijing University of Posts and Telecommunications · 2017–<br>B.Eng. in Internet of Things Engineering | 2021                    | University of Warwick · M.Sc. in Cyber Security<br>Engineering | 2021–<br>2022 |
| Xi'an Jiaotong-Liverpool University · Ph.D.<br>Candidate, School of Intelligent Engineering            | 2023–2027<br>(expected) |                                                                |               |

## Research Profile

Ph.D. candidate at the School of Intelligent Engineering, Xi'an Jiaotong-Liverpool University, focusing on trustworthy generative audio, music intelligence, and AI security. My methodological focus is **privacy auditing, adversarial generation, and perceptually aligned evaluation**. I propose and validate methods for membership inference, music diffusion, and symbolic-music models; as a project lead, I also advance MusicXML-to-audio rendering research into a creator-facing proof of concept.

## Research and Project Experience

Innovative methods → creator-facing delivery

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|--------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| <b>2020–2021 · Audio Algorithm Foundations</b><br>Contributed to singing voice generation and enhancement research at Changba. | <b>2023–2027 · Ph.D. Research</b><br>Propose and validate generative-audio security methods. | <b>Mar. 2027 · Ph.D. Graduation</b><br>Available for AI, audio intelligence, and trustworthy-model roles. |
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### Trustworthy Generative Audio and Music Intelligence Ph.D. Research · Researcher

|                |                                                                                                                                                                                   |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Role</b>    | Membership inference determines whether a work was used in model training, making it important for auditing data-privacy and copyright risks in generative models.                |
| <b>Action</b>  | Develop membership-inference and auditing methods for music diffusion and symbolic-music generation models, with emphasis on reliable identification at low false-positive rates. |
| <b>Outcome</b> | Produced <b>11 publications and workshop presentations</b> across ICASSP, INTERSPEECH, ISMIR, AAAI Workshop, CMMR, FG, and DCASE.                                                 |

### Music Adversarial Attacks and Perceptual Evaluation Ph.D. Research · First / Co-first Author · 2025–2026

|                     |                                                                                                                                                                                                            |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Role</b>         | Addressed the difficulty of jointly achieving attack effectiveness and human imperceptibility in music retrieval, genre classification, and cover-song identification.                                     |
| <b>Action</b>       | Proposed MAIA: locate critical audio segments and use generative inpainting for white-box and black-box attacks; further proposed the psychoacoustically conditioned PAMT perceptual representation model. |
| <b>Contribution</b> | MAIA achieved high attack success and audio fidelity across MIR tasks; PAMT reached <b>0.65</b> correlation with subjective listening scores and improved robust accuracy by <b>9.15%</b> on average.      |

### Parameterized Score-to-Audio Expressive Rendering for Creators Suzhou Embodied Intelligence Future Education

Concept Verification Center PoC · Principal Investigator · Sep. 2025–Aug. 2026

|                |                                                                                                                                                                                                         |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Role</b>    | Led the Suzhou Embodied Intelligence Future Education Concept Verification Center project as <b>Principal Investigator (PI)</b> , accountable for end-to-end progress, delivery, and validation.        |
| <b>Action</b>  | Coordinated expressive-rendering models, macOS application requirements and integration, evaluation design, and user trials; defined import, A/B comparison, auditioning, export, and preset workflows. |
| <b>Outcome</b> | A single score produces multiple performance styles; the prototype reached <b>TRL 5</b> in small-scale relevant-setting validation.                                                                     |

Changba Technology Co., Ltd.    Audio Processing Intern · Nov. 2020–Jun. 2021

|         |                                                                                                                                                           |
|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Role    | Contributed to singing-voice generation and enhancement research for online karaoke applications, covering audio processing and algorithm validation.     |
| Action  | Conducted audio simulation, algorithm testing, and evaluation to support research iteration in singing voice generation and enhancement.                  |
| Outcome | Built practical experience from audio algorithms to evaluation in real product settings, providing an engineering foundation for later audio-AI research. |

Selected Publications

First / co-first author first · Full list on Google Scholar

|                    |                                                                                                                                                                                                                                                                        |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ICASSP 2026        | Yuxuan Liu et al. <i>Membership Inference Attack Against Music Diffusion Models via Generative Manifold Perturbation</i> .<br>IEEE International Conference on Acoustics, Speech and Signal Processing                                                                 |
| ISMIR 2025         | Yuxuan Liu et al. <i>MAIA: An Inpainting-Based Approach for Music Adversarial Attacks</i> .<br>International Society for Music Information Retrieval Conference                                                                                                        |
| AAAI Workshop 2026 | Yuxuan Liu et al. <i>TS-RaMIA: Membership Inference Attacks for Symbolic Music Generation Models</i> . PMLR 303:1–15.<br>AAAI Conference on Artificial Intelligence Workshop                                                                                           |
| CMMR 2025          | Yuxuan Liu et al. <i>Training a Perceptual Model for Evaluating Auditory Similarity in Music Adversarial Attack</i> .<br>International Symposium on Computer Music Multidisciplinary Research                                                                          |
| INTERSPEECH 2026   | P. Zhang, Z. Li, Yuxuan Liu, Y. Cai, Y. Tan, S. Li. <i>CoRE: Contrastive Evidence-Aware Rescoring for Multiple-Choice Audio Question Answering</i> .<br>Annual Conference of the International Speech Communication Association                                        |
| INTERSPEECH 2026   | Y. Cai, Y. Tan, P. Zhang, Yuxuan Liu, S. Li, X. Shao. <i>Enhancing Temporal Prediction Consistency for Short-Duration Acoustic Scene Classification via Semantic Adversarial Training</i> .<br>Annual Conference of the International Speech Communication Association |
| ICASSP 2026        | P. Zhang, Yuxuan Liu et al. <i>DDSC: Dynamic Dual-Signal Curriculum for Data-Efficient Acoustic Scene Classification under Domain Shift</i> .<br>IEEE International Conference on Acoustics, Speech and Signal Processing                                              |
| ICASSP 2026        | P. Zhang et al. <i>TopSeg: A Multi-Scale Topological Framework for Data-Efficient Heart Sound Segmentation</i> .<br>IEEE International Conference on Acoustics, Speech and Signal Processing                                                                           |
| FG 2026            | P. Zhang, Yuxuan Liu, Z. Li, R. Sang, S. Li. <i>LBOR: Laplace–Beltrami Operator Regularization for Robust Skeleton-based Isolated Sign Language Recognition</i> . pp. 1–5.<br>IEEE International Conference on Automatic Face and Gesture Recognition                  |
| DCASE 2025         | P. Zhang (co-first author), Yuxuan Liu (co-first author) et al. <i>An Entropy-Guided Curriculum Learning Strategy for Data-Efficient Acoustic Scene Classification under Domain Shift</i> .<br>Workshop on Detection and Classification of Acoustic Scenes and Events  |
| AAAI Workshop 2026 | R. Sang, Yuxuan Liu. <i>SceneGuard: Training-Time Voice Protection with Scene-Consistent Audible Background Noise</i> .<br>AAAI Conference on Artificial Intelligence Workshop                                                                                         |

Core Competencies

|              |                                                                                                            |
|--------------|------------------------------------------------------------------------------------------------------------|
| Generative   |                                                                                                            |
| Audio        | Diffusion models, music/speech generation, symbolic music modeling, and MusicXML workflows                 |
| Trustworthy  |                                                                                                            |
| AI           | Membership inference, model privacy, copyright auditing, provenance, and adversarial robustness evaluation |
| Audio        |                                                                                                            |
| Intelligence | MIR, audio signal processing, acoustic scene classification, and perceptually aligned evaluation           |

|                        |                                                                                                                                 |
|------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| Agent                  |                                                                                                                                 |
| Toolchain              | Codex, Cursor, and Claude Code for implementation, debugging, research-prototype iteration, and document automation             |
| Engineering & Research | Python, C/C++ , MATLAB, Java, and SQL; experimental design, statistical analysis, academic writing, and user-trial organisation |